

Composition Collapse: Stable Factual Knowledge Does Not Imply Compositional Reasoning

Anonymous ACL submission

Abstract

Post-training is routinely evaluated through aggregate benchmark scores that treat multi-hop reasoning as a single capability—as if a model that answers more questions correctly must be better at assembling facts. We show that this assumption can be misleading: recipes with statistically indistinguishable atomic knowledge produce composition behaviour separated by over 40 percentage points, a phenomenon we call *composition collapse*: the systematic failure to assemble stably-known facts into chains, invisible to aggregate metrics. We introduce a double-gate protocol that changes the estimand from an aggregate compositionality gap to residual composition failure conditioned on stable atomic access, decomposing post-training gains into three independent channels: atomic stability, residual composition, and critical depth. On a benchmark of temporal factual chains spanning depths 2–11 across four post-training recipes, this decomposition reveals that post-training objectives shift composition capability in directions that aggregate metrics mask, and suggests that claims about multi-hop reasoning improvement should be accompanied by atomic-gate-controlled composition metrics. Diagnostic probes further show that a substantial share of measured composition failure reflects generation-time computation constraints rather than permanent inability to compose.

1 Introduction

When a language model fails a multi-hop question—*which came first, the telephone or flight?*—two explanations are possible. Either the model does not reliably know the dates, or it knows both and still cannot compare them. These demand different fixes: more factual knowledge versus better compositional reasoning. Yet standard evaluation practice treats them as the same thing.

The field evaluates multi-hop reasoning through aggregate benchmark accuracy on datasets like HotpotQA (Yang et al., 2018): a model that answers

more composite questions correctly is assumed to be better at composing facts. The *compositionality gap* (Press et al., 2023; Dziri et al., 2023)—the drop from single-hop to multi-hop accuracy—is widely used to quantify composition failure. But this metric conflates two categorically different failure modes. A model may answer a sub-question correctly once without *stable* access to the underlying fact, yet a single correct answer does not guarantee the model can reproduce that knowledge in a longer chain. Attributing the entire gap to “composition failure” mixes atomic-instability variance into what is claimed as a reasoning deficit. Prior work has shown that knowledge extraction can fail systematically even when facts are present in the training distribution (Allen-Zhu and Li, 2023), and that multi-hop failures can be localised to specific layer-level timing problems (Biran et al., 2024). Recent work on shortcut-free latent multi-hop evaluation (SOCRATES; Yang et al. (2025b)) shows that composition success varies dramatically by entity type even when atomic facts are known—reinforcing the need to measure composition net of atom confidence. These studies establish that reasoning failures exist independently of memorisation—a premise we share. But they do not provide a measurement protocol that cleanly separates the two failure modes at evaluation time.

This conflation is not a minor measurement issue. Two post-training recipes whose atomic stability differs by less than two percentage points can diverge by over 40 pp in residual composition failure at the shallowest non-trivial depth. We call this *composition collapse*: the systematic failure to assemble stably-known facts into correct chains, hidden from aggregate metrics and visible only when atomic knowledge is controlled for.

We introduce a *double-gate protocol* that changes the estimand from an aggregate gap to residual composition failure conditioned on stable atomic access. Before measuring whether a model

can compose facts, the protocol first verifies that it stably possesses each fact (consistency across paraphrases) and can answer each sub-question in isolation. Errors that survive both gates are *residual composition failure*: the model stably knows the parts, has verbalised them one at a time, and still fails to assemble them. With this protocol, post-training gains decompose into three independent channels: Δ_{atom} (atomic stability), Δ_{comp} (residual composition at matched atoms), and Δ_{depth} (the depth at which residual failure exceeds 50%). We instantiate the protocol on D4v2, a 390-question temporal benchmark spanning four task families at depths 2–11.

The benchmark covers temporal factual chains (dates, events, causal order); domain generality beyond temporal reasoning remains open and is examined preliminarily through in-context probes and a cross-domain pilot (§4).

We evaluate six open-weights models at 7–13B scale spanning four post-training recipes—base, RLHF (Ouyang et al., 2022), SFT reasoning-trace distillation, and native outcome-verified RL—and run a controlled same-base Hu et al. (2022) LoRA intervention (SFT-answer, SFT-trace, Shao et al. (2024) GRPO) that isolates the training objective from base-model confounds. Three findings emerge. First, base models are filtered by atomic knowledge, not composition—their atomic gate admits <5 cases at any depth. Second, at matched atomic stability, residual failure ranges from 0% to 47% at depth 2 depending on the training objective, and outcome-verified RL outperforms SFT distillation by a wide margin under identical data and compute. Third, training on a composition depth closes most of that depth’s residual failure gap but transfers only weakly to adjacent held-out depths. Enabling chain-of-thought reasoning reduces residual failure by $3\text{--}5\times$ (§6), locating much of the bottleneck in generation-time computation, not in static knowledge representation. Our contributions are: (i) a double-gate protocol that provides a cleaner measurement of composition net of atomic knowledge; (ii) a three-channel decomposition that makes post-training recipes comparable on a like-for-like basis; and (iii) a controlled causal intervention isolating a post-training-specific composition channel with limited depth transfer. We do not claim to reveal the essence of post-training; rather, we offer a measurement protocol that exposes previously hidden variation across recipes at matched atoms.

2 Related Work

Measuring the compositionality gap. Press et al. (2023) formalised the compositionality gap as the drop from single-hop to multi-hop accuracy; Dziri et al. (2023) showed Transformer performance on multi-hop composition degrades with chain length. Yang et al. (2025b) introduced latent multi-hop probes (SOCRATES) that bypass surface shortcuts, showing composition success varies dramatically by entity type even when atomic facts are known. These frameworks share a confound: the single-hop baseline does not verify that facts are *stably* accessible—a single correct sub-question answer does not guarantee the model can reproduce that knowledge when the same fact appears embedded in a chain. Our double-gate protocol adds a per-fact stability filter that separates transient retrieval from stable knowledge, converting a population-level correction into a per-case admissibility test.

Post-training and reasoning. RLHF (Ouyang et al., 2022), SFT reasoning-trace distillation (Wei et al., 2022; DeepSeek-AI et al., 2025), and outcome-verified RL (Shao et al., 2024) are the dominant paradigms for improving reasoning. Prior work evaluates these recipes through aggregate benchmark scores, which conflate knowledge acquisition with reasoning improvement. Our three-channel decomposition shows these recipes shift atomic stability and residual composition independently: atomic stability saturates after the base-to-instruct jump, while residual composition varies by 40+ pp across recipes at matched atoms. This decomposition is orthogonal to the training objective: it provides a measurement lens, not a modelling claim.

Computation and composition. Biran et al. (2024) localised multi-hop failure to layer-level timing—the model encodes both facts but processes the second too late. Allen-Zhu and Li (2023) showed knowledge extraction fails systematically even when facts are in the training distribution, and CoT data during training can bridge the gap. Our CoT diagnostic (§6) complements these findings: enabling CoT at inference time recovers 70–75% of gate-passing failures, locating much of the residual failure in generation-time computation rather than permanent representational deficit. Together with the patching experiment (Appendix I), which finds negligible information in prompt-end representations, these results triangulate the bottleneck to the generation process itself.

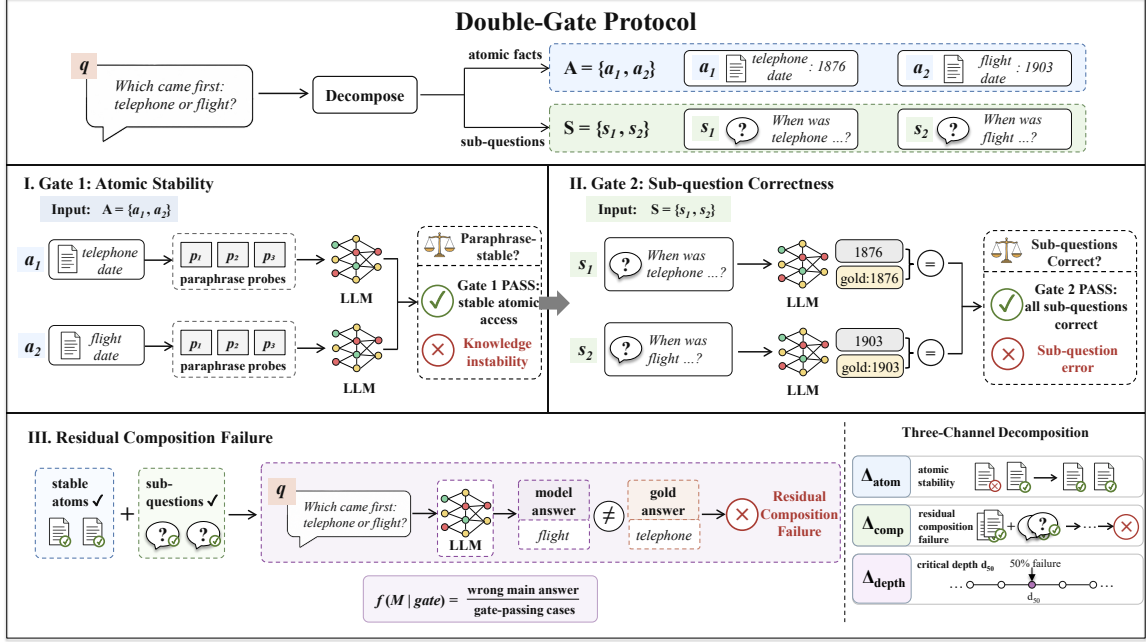


Figure 1: **Double-gate protocol for measuring residual composition failure.** A multi-hop question is decomposed into atomic facts and aligned sub-questions. **I.** Atomic Stability verifies stable and correct access to each atom across paraphrases. **II.** Sub-question Correctness verifies that each sub-question is answered correctly in isolation. **III.** Residual Composition Failure measures whether the original multi-hop question is still answered incorrectly after both gates are passed. This separates composition errors from unstable knowledge access and supports the three-channel decomposition into Δ_{atom} , Δ_{comp} , and Δ_{depth} .

3 Method

Figure 1 summarises the pipeline. A composition instance is a main question q whose answer requires k atomic facts $a_1, \dots, a_k$ (the depth). Each atom carries m_i paraphrased probes plus a sub-question s_i matching how the atom appears in the chain. An instance *passes the atomic gate* iff every atom is answered consistently across all paraphrases (consistency matcher of Press et al., 2023); it *passes the sub-question gate* iff every s_i is correct in isolation. Our primary quantity is the *residual composition failure rate*: the fraction of gate-passing instances whose main answer is still wrong. By construction it upper-bounds pure composition failure—format artifacts, distractor attention, and greedy-trajectory degeneracy can inflate the measured rate without reflecting genuine compositional inability (we quantify this margin in §7).

Given a recipe R and a matched comparator R' (atomic stabilities within 2 pp), we decompose the post-training benefit into three channels:

$$\begin{aligned}\Delta_{\text{atom}}(R) &= s(M_R) - s(M_{R'}), \\ \Delta_{\text{comp}}(R) &= f(M_R | \text{gate}) - f(M_{R'} | \text{gate}), \\ \Delta_{\text{depth}}(R) &= d_{50}(M_R) - d_{50}(M_{R'}),\end{aligned}$$

where s is atom stability, f is residual composition failure, and d_{50} is the depth at which f first exceeds 50%. When atomic stabilities are matched, Δ_{comp} isolates a pure composition benefit; the $\Delta_{\text{comp}} - \Delta_{\text{depth}}$ gap later lets us distinguish a training-data effect from a model-structure effect.

Benchmark. D4v2 contains 390 main questions and 2490 atomic sub-questions built from four task families: a depth-2 pair-ordering control and three depth-covering families (TEMPORAL_RANK, TEMPORAL_SUCCESOR, TEMPORAL_INTERVAL_DECOY) at $d \in \{4, 6, 8\}$, with TEMPORAL_INTERVAL_DECOY additionally sampling intermediate fact counts ($d=7, 9, 11$; $n=30$ each). Task families differ in required chain structure—ranking a set, stepping forward in time, and disambiguating overlapping intervals—so same-depth cases are not interchangeable in difficulty. To enable cross-family comparison at shared depth bins, we group depth-7 cases with depth 4, depth 9 with depth 6, and depth 11 with depth 8 (see Appendix B for rationale and full breakdown). Three additional synthetic task families (KINSHIP, NUMERICAL, SPATIAL) built from fictional entities form a separate evaluation suite; as discussed in §4, they are excluded because our closed-book pro-

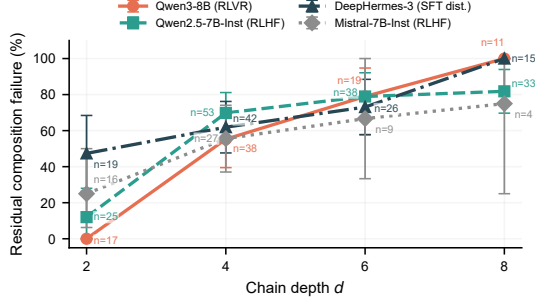


Figure 2: **The composition collapse curve.** Residual composition failure rate versus depth on D4v2 (bootstrap 95% CI). All gate-passing recipes cross 50% failure by depth 4–5; two reach 100% by depth 8. Base models are omitted (see Appendix C); their atomic gate admits <5 cases at every depth. CI width grows with depth as gate-passing n declines; points at depth 8 should be interpreted with particular caution ($n \leq 37$ for all recipes).

protocol cannot evaluate composition when internal atomic knowledge is absent.

Recipe coverage and evaluation. We evaluate four post-training recipes: base, RLHF, SFT distillation of reasoning traces, and native outcome-verified RL (RLVR), across six open-weights models at the 7–13B scale (checkpoints in Appendix C). All numbers use the Press et al. consistency matcher; full adjudication tiers and vLLM vs. HuggingFace cross-validation are in Appendix K and H. Inference is greedy ($T=0$), XML-structured.

4 Experiments and Results

We report three observations that rule out the most obvious alternative explanations and isolate a post-training-specific composition channel. First, the atomic gate removes base models entirely, so the failures we measure are not knowledge gaps (§4, paragraph 1). Second, two recipes with statistically indistinguishable atomic knowledge produce categorically different composition at the shallowest non-trivial depth (paragraph 2). Third, within a fixed depth, task structure changes residual failure by up to 66 percentage points, so depth is not the dominant axis of difficulty (paragraph 3).

Figure 2 shows residual composition failure as a function of chain depth, with bootstrap 95% CIs over atomic-gate passing cases.

Base models are filtered by knowledge, not composition. All three pre-RLHF base models we test (Llama-3-8B (Grattafiori et al., 2024), Mistral-

Table 1: Residual composition failure rate (%) by depth on D4v2, restricted to gate-passing post-trained models. “Atom stab.” is the per-probe atomic-stability rate; the atomic gate imposes a stricter per-fact criterion, so the effective fact-level pass rate is lower. Parenthesised values are n , the number of atomic-gate-passing cases. We omit $n < 5$.

Model	Stab. (%)	$d=2$	$d=4$	$d=6$	$d=8$
Qwen3-8B (Yang et al., 2025a) (RLVR)	87.7	0.0 (17)	55.3 (38)	78.9 (19)	100.0 (11) [†]
Qwen2.5-7B-Inst (Qwen et al., 2024) (RLHF)	89.9	12.0 (25)	69.8 (53)	78.9 (38)	81.1 (37)
DeepHermes-3 (Teknium et al., 2025) (SFT distill)	88.0	47.4 (19)	61.9 (42)	73.1 (26)	100.0 (15) [†]
Mistral-7B-Inst (Jiang et al., 2023) (RLHF)	78.8	25.0 (16)	55.6 (27)	66.7 (9) [†]	–

[†] $n < 15$ gate-passing cases; point estimates in these cells should be interpreted with caution—bootstrap CIs exceed ± 25 pp.

7B-v0.1 (Jiang et al., 2023), Qwen2.5-7B (Qwen et al., 2024)) have atomic stability below 15%, so their joint stability at depth k is near zero and the atomic gate admits fewer than five cases at any depth. Attributing their failure to “composition” would be a category error—precisely the confound our protocol is designed to remove. Full per-model atomic rates are reported in Appendix C.

Synthetic task families are excluded by design.

Three synthetic families (KINSHIP, NUMERICAL, SPATIAL) built from fictional entities are part of the full suite but are excluded because no model can possess internal knowledge of entities absent from its training data (atomic accuracy = 0%); the protocol correctly prevents composition measurement when atoms are absent. We therefore restrict analysis to temporal facts where models demonstrably possess the requisite knowledge.

4.0.1 Composition collapse persists under in-context evidence

To verify that composition difficulty is not merely an artefact of closed-book knowledge access, we evaluated Qwen2.5-7B-Instruct on the same synthetic families with all facts supplied in the prompt as [Evidence] blocks. Under this in-context protocol, only 21.8% of cases survive the sub-question gate, and on those gate-passing cases residual composition failure is 88.9% (Table 2). This convergence—88.9% in-context versus 55–100% closed-book at matched depths—rules out knowledge retrieval as the dominant explanation for composition collapse. When the relevant facts are supplied in the prompt and gate-passing cases are conditioned on correct sub-question use, composition remains the primary failure mode; the bottleneck is assembly, not lookup.

Table 2: In-context composition on synthetic families (Qwen2.5-7B-Instruct). Facts are supplied in the prompt. “Atom pass” = all sub-questions correct. Residual failure = main-question error given atom pass.

Task family	n	Atom pass	Residual failure (%)
Kinship (matrilineal)	33	8	62.5
Kinship (patrilineal)	33	9	66.7
Spatial (eastward)	33	12	91.7
Spatial (northward)	33	6	83.3
Numerical (offset)	33	1	100.0
Aggregate	165	36	88.9

Quantifying the conflation. We recompute residual failure using only the sub-question gate (the standard single-gate protocol). The aggregate single-gate rate exceeds the double-gate rate by 2.5–9.8 pp across models, but the cell-level inflation peaks at 11.4 pp, and the atomic gate removes 15–44% of single-gate-passing cases—cases prior work would count as valid composition trials whose constituent atoms are not stably known. These findings motivate the double-gate protocol not as a refinement but as a correction of the estimand.

Matched atoms, divergent composition: the composition collapse. Three recipes admitted by the gate cluster tightly on atomic stability—Qwen3 87.7%, DeepHermes 88.0%, Qwen2.5-Instruct 89.9%—yet collapse to 0%, 47%, and 12% residual composition failure at depth 2 (Table 1). This *composition collapse*—a 47-point gap between a native outcome-verified RL model and a recipe that merely imitates its reasoning traces—is the paper’s cleanest correlational signal: at the knowledge level these recipes are indistinguishable, at the composition level they are categorically different.

Limited depth transfer under current recipes. Table 1 shows that all four gate-passing recipes sit at or above 67% residual failure by depth 6, and every reportable measurement at depth 8 is 75% or higher; Qwen3 and DeepHermes both reach 100%. The best recipe we evaluate pushes d_{50} no higher than 5–6 hops. Within our recipe coverage—two RLHF variants, an SFT reasoning-trace distillation, and a native RLVR model—the depth ceiling does not meaningfully move, suggesting that current post-training recipes improve shallow composition more reliably than depth extrapolation.

Structure dominates depth. At fixed depth 4, Qwen3’s residual failure ranges from 38.5% on

Table 3: Residual failure (%) by task family and depth for Qwen3-8B (double gate). At fixed depth, task structure changes failure by up to 54 pp. Parenthesised values are n . “–” = $n < 5$.

Task family	$d=2$	$d=4$	$d=6$	$d=8$
Pair control	0.0 (17)	–	–	–
Temp. successor	–	38.5 (13)	88.9 (9)	–
Temp. int. decoy	–	41.2 (17)	57.1 (7)	80.0 (5)
Temp. rank	–	92.9 (14)	83.3 (6)	100.0 (6)

TEMPORAL_SUCCESSION to 92.9% on TEMPORAL_RANK—a 54-point gap at identical chain length (Table 3). The within-depth spread even exceeds the within-task depth progression on TEMPORAL_INTERVAL_DECOY (41% at $d=4 \rightarrow 57\%$ at $d=6 \rightarrow 80\%$ at $d=8$). The ordering of models by residual failure reverses across task families, which is incompatible with any account that treats depth as a monotone difficulty axis.

4.1 Decomposition into three channels

Having established that composition failure survives the atomic gate and varies dramatically across recipes, we decompose post-training benefits into the three channels of §3: atomic stability (Δ_{atom}), residual composition accuracy at matched atoms (Δ_{comp}), and critical depth (Δ_{depth}).

4.1.1 Δ_{atom} : post-training buys atomic stability in large steps

The base-to-instruct jump is 76–88 pp of atomic stability; once paid, further improvements are single percentage points. For a 7–13B model, “does post-training help composition?” is first a question about whether the atoms ever become stable enough to attempt composition.

4.1.2 Δ_{comp} : at matched atoms, recipe matters by 40+pp

At matched atomic stability (87–90%), the three gate-passing recipes diverge by 47 pp at depth 2 (Table 1): taking Qwen3 (RLVR) as reference, Δ_{comp} is –47.4 pp (DeepHermes) and –12.0 pp (Qwen2.5-Inst). Two recipes indistinguishable on knowledge benchmarks differ by 35 pp in residual composition. The collapse of SFT-distilled reasoning suggests that distilling reasoning traces can induce surface pattern-matching that generalises worse than outcome-verified RL. Base-model confounds for DeepHermes (Llama-3 vs. Qwen2) are partly addressed via R1-distill comparisons in Appendix C.

Table 4: Critical depth (d_{50}) per task family, consistency gate. d_{50} is estimated by linear interpolation between the two depth bins ($\{2, 4, 6, 8\}$) bracketing 50% residual failure. Ranges reflect the uncertainty band from the 95% Clopper–Pearson CI on the underlying binomial rates. All values are approximate due to the coarse 4-bin resolution and small gate-passing sample sizes at deeper depths.

Model	RANK	SUCC.	INT_DECOY	Agg.
Qwen3 (RLVR)	<4	[4,6]	[4,6]	[3,5]
Qwen2.5-Inst (RLHF)	<4	[4,6]	[4,6]	[3,5]
DeepHermes (SFT)	<4	<4	<4	<3
Mistral-Inst (RLHF)	<4	<4	[4,5]	<3

4.1.3 Δ_{depth} : shallow depth generalisation

Across post-trained models, d_{50} averages in $[3, 5]$ (Table 4). Even the strongest recipe does not exceed $d_{50}=5-6$, and TEMPORAL_RANK pushes d_{50} below 4 for every model. Post-training improves shallow composition more reliably than depth extrapolation.

5 Causal intervention: training reduces seen-depth collapse with limited transfer

The cross-recipe analysis in §4 is correlational. To rule out confounding by base model, we train four controlled LoRA-GRPO variants on Qwen2.5-7B-Instruct (Qwen et al., 2024)¹: v1 trains on depths $\{2, 4\}$, v2 on $\{2, 4, 6\}$, v3 on $\{2, 4, 6\}$ with heavier budget (rank 64, 8 epochs), and v4 extends the heavy-budget setup to train on $\{2, 4, 6, 8\}$. v1 holds out depths $\{6, 8\}$ at test time; v2 and v3 hold out $\{8\}$; v4, which trains on all four D4V2 depths $\{2, 4, 6, 8\}$, evaluates depth 8 in-distribution. Evaluation uses the same protocol as §4.

Four causal conclusions. (1) *RL on composition improves composition net of atoms.* Training cuts depth-4 residual failure by 45–62 pp across variants while atomic stability moves by 1–3 pp; the effect is a clean Δ_{comp} . At depths 4 and 6, baseline and GRPO v1 bootstrap CIs are disjoint (Figure 3). (2) *OOD depth generalisation decays.* Training on $\{2, 4\}$ generalises 20 pp to depth 6, but only 7 pp to depth 8. (3) *In-distribution gains do not transfer to deeper hold-out depths—unless the deeper*

¹All variants use learning rate 10^{-5} , 8 generations per prompt, reward = consistency match + XML format + anti-template-echo, Unsloth (Han et al., 2023) (memory-efficient LoRA (Hu et al., 2022)) with patched TRL. v1/v2: LoRA rank 32, 4 epochs, ~ 1 h. v3/v4: LoRA rank 64, 8 epochs, ~ 3.2 h (v3) / ~ 5 h (v4). Details in Appendix F.

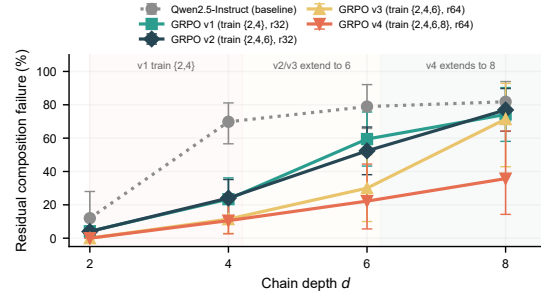


Figure 3: Baseline (Qwen2.5-Instruct, dotted) vs. four LoRA-GRPO variants. Shaded bands mark training-depth regions. v1/v2 (rank 32, 4 epochs) show that adding depth-6 training improves $d=6$ but not $d=8$; v3 (rank 64, 8 epochs) further reduces $d=6$ to 27.9% with minimal $d=8$ transfer; v4 (train $\{2, 4, 6, 8\}$, rank 64, 8 epochs) cuts $d=8$ to 36.0%, demonstrating that the depth ceiling moves with training-data coverage. All CIs are bootstrap 95%.

Table 5: Residual failure (%) for four GRPO variants vs. baseline. v2 adds depth-6 to training (7 pp improvement at $d=6$); v3 uses heavier budget (rank 64, 8 epochs), cutting $d=4$ and $d=6$ residual failure by an additional 15–24 pp. v4 (train $\{2, 4, 6, 8\}$, rank 64, 8 epochs) reduces $d=8$ by 45 pp vs. baseline and by 36 pp vs. v3, showing that training on depth 8 largely closes the depth-8 gap. Atomic stability remains within 1 pp of baseline (89.9%) for all four.

	$d=2$	$d=4$	$d=6$	$d=8$
Qwen2.5-Inst baseline	12.0	69.8	78.9	81.1
GRPO v1 (train $\{2, 4\}$, r32)	4.2	24.5	59.5	74.2
GRPO v2 (train $\{2, 4, 6\}$, r32)	4.0	24.1	52.4	76.9
GRPO v3 (train $\{2, 4, 6\}$, r64)	0.0	8.5	27.9	71.9
GRPO v4 (train $\{2, 4, 6, 8\}$, r64)	0.0	7.8	20.6	36.0
Δ v1	−7.8	−45.3	−19.5	−6.9
Δ v2	−8.0	−45.7	−26.6	−4.2
Δ v3	−12.0	−61.3	−51.0	−9.2
Δ v4	−12.0	−62.0	−58.3	−45.1

depth is included in training. Adding depth-6 to training (v2) improves depth 6 by 7 pp but does not shift depth 8 relative to v1. Heavier budget without depth-8 data (v3) cuts $d=6$ to 27.9% but leaves $d=8$ at 71.9%, only 5 pp below v2. The ratio of out-of-distribution gain to in-distribution gain is 0.21, showing that deeper training data alone does not transfer to farther hold-out depths under a LoRA-GRPO setup. (4) *Depth-8 failure is trainable when depth-8 data is included.* v4 (train $\{2, 4, 6, 8\}$, rank 64, 8 epochs) reduces $d=8$ residual failure to 36.0%—a 45 pp drop from baseline and a 36 pp drop from v3. All other depths improve modestly (e.g., $d=6$ drops from 27.9% to 20.6%). Because v4 evaluates depth 8 in-distribution (depth 8 is in its training mix), this gain reflects training-coverage

Table 6: Same GRPO recipe (train $\{2, 4, 6\}$, rank 32, 4 epochs) on two base-model families. Δ_{comp} direction and magnitude replicate. Parenthesised values are n (double-gate-passing cases).

Model	Residual failure (%)		
	$d=2$	$d=4$	$d=6$
Qwen2.5-Inst (baseline)	12.0 (25)	66.7 (66)	78.7 (47)
Qwen2.5-Inst + GRPO v2	4.0 (25)	30.8 (65)	58.2 (55)
Llama3-8B-Inst + GRPO	0.0 (19)	17.9 (39)	42.9 (28)

extension rather than generalised depth-8 reasoning capability; it demonstrates that the ceiling is governed by whether a depth appears in training, not by a hard architectural limit. Across all four variants, the clearest signal is that *training on a depth closes most of that depth’s residual failure gap, but transfers only weakly to adjacent depths.*

Cross-model replication. The same GRPO recipe (train $\{2, 4, 6\}$, LoRA rank 32, 4 epochs) applied to Llama3-8B-Instruct (Grattafiori et al., 2024) reduces $d=4$ residual failure to 17.9% and $d=6$ to 42.9%—outperforming the matched Qwen GRPO v2 by 13 pp at $d=4$ and 15 pp at $d=6$ (Table 6). The Δ_{comp} direction and magnitude replicate across base-model families, ruling out a Qwen-specific artifact. On Mistral-7B-v0.1 (Jiang et al., 2023) (base), the same GRPO recipe lifts atomic stability from $<15\%$ to a level where 34 cases survive the double gate at $d=4$, with 0% residual failure—suggesting that GRPO can simultaneously improve atomic knowledge and composition when starting from a low base, though the small gate-passing n warrants caution.

Training objective vs. base model. To isolate the training objective from depth, we trained two additional variants on the same base model (Qwen2.5-7B-Instruct), same data ($d \in \{2, 4\}$), and same LoRA budget: SFT-answer (gold answer only) and SFT-trace (reasoning chain + answer). Under the double gate, GRPO’s residual failure at depth 4 (24.5%) is 52 pp lower than SFT-trace (76.9%) and 39 pp lower than SFT-answer (63.8%; Table 7). SFT-trace is *worse* than the untrained baseline (69.8%), confirming that distilling reasoning traces can actively damage composition. SFT-trace also performs *worse* than SFT-answer, which learns only the final answer—under a clean gate, imitating reasoning chains does not reliably transfer to novel composition questions.

Table 7: Same base model, same data ($d \in \{2, 4\}$), same LoRA budget. Under the double gate, GRPO cuts residual failure by 39–52 pp relative to SFT.

Recipe	Atom (%)	$d=2$	$d=4$
Baseline (untrained)	88.6	12.0	69.8
SFT-answer (answer only)	85.3	19.0	63.8
SFT-trace (chain + answer)	85.8	23.5	76.9
GRPO v1 (outcome RL)	87.1	4.2	24.5

6 Diagnostics

We apply two diagnostics that constrain the space of alternative explanations for residual failures. Additional probes (prompt-end patching, contamination control, hidden-state trajectory) are in Appendix I and J.

Thinking mode as a generation-process probe.

If residual failures originate from insufficient generation-time computation, allocating more inference compute should reduce the failure rate. We evaluated Qwen3-8B on D4V2 with native CoT reasoning enabled (thinking=true, up to 16K internal tokens). Under thinking, atomic stability is 82.3% (vs. 87.7% standard). Critically, 70–75% of gate-passing failures are recovered: 72.2% at $d=4$ (13/18 failures fixed), 70.0% at $d=6$ (7/10), 75.0% at $d=8$ (3/4). After thinking, residual failure is nearly flat across depth—28.6%, 18.8%, 23.1% at depths 4,6,8—compared to the steep 56.8%→92.3% curve in standard mode. The collapse curve largely disappears under more generation-time computation, locating the bottleneck in the inference process. By task family, TEMPORAL_RANK drops to 4.0% under thinking while TEMPORAL_SUCCESSOR remains at 53.1%, indicating certain chain structures resist even extended CoT.

Convergent evidence from explicit CoT. On Qwen2.5-7B-Instruct with explicit CoT prompting, atomic stability drops 10 pp (to 79.7%) but residual failure falls from ~ 70 –75% to 43.2% overall, with the depth gradient flattening: 39.3% at $d=4$, 68.8% at $d=6$, 55.6% at $d=8$ (Appendix S). That explicit reasoning on a different model family produces the same qualitative reduction—while atomic stability moves oppositely—confirms the generation process as the locus of residual failure.

Implication. The CoT results locate residual failure in the generation process: the majority of *composition collapse* reflects a generation-time compu-

tation constraint. The causal GRPO result (§5)—that RL on composition training reduces residual failure net of atomic stability, replicating across model families—shows the effect is trainable.

Failure taxonomy. To bound how much format artifacts inflate the residual failure rate, we adjudicated all 251 double-gate failure candidates across models and task families (Appendix O). 72% (181/251) are structural composition failures—temporal boundary confusion, intermediate variable loss, binding shifts—while 28% (70/251) are scoring or format artefacts where the model’s reasoning is substantially correct. This bounds the non-compositional inflation of residual failure at roughly 14–21% of gate-passing errors, confirming that our primary metric primarily reflects genuine composition inability rather than measurement noise.

7 Discussion

7.1 Synthesis

The central lesson is that “post-training” is not a single knob: it affects atomic knowledge and composition through distinct, separable channels. Two recipes with indistinguishable atomic knowledge produce composition behaviour separated by 40+ pp.

The strongest empirical contrast is between SFT distillation and native outcome-verified RL. Distillation supplies format but no grounding incentive; outcome-verified RL rewards only the final token. Under identical base model, data, and LoRA budget, GRPO outperforms SFT-trace by 52 pp at depth 4 (Table 7); SFT-trace is *worse* than the untrained baseline, confirming that distilling reasoning chains can actively damage composition. The Δ_{comp} direction replicates across base-model families: the same GRPO recipe on Llama3-8B-Instruct yields 17.9% residual failure at $d=4$ (Table 6). We do *not* claim SFT distillation learns only superficial imitation—only that reasoning-chain distillation did not reliably translate into residual composition ability.

The critical-depth result shows that training on a depth closes most of that depth’s gap but transfers only weakly to adjacent depths (Table 5): adding depth-6 to training leaves $d=8$ unchanged, and when depth-8 is included, $d=8$ drops from 81.1% to 36.0%, showing the ceiling tracks data coverage rather than fixed capacity. Task structure changes

residual failure by up to 54 pp at fixed depth (Table 3), precluding any single “depth ceiling” number.

Three diagnostics triangulate the bottleneck. The collapse persists in an in-context setting (88.9% residual failure; Table 2), ruling out retrieval failure. 70–75% of gate-passing failures are recovered by chain-of-thought prompting (Wei et al., 2022) (§6), locating the bottleneck in inference rather than static representation. The three-channel decomposition is thus a simple correction for making post-training comparisons more interpretable.

A cross-domain pilot (Appendix L) confirms the phenomenon is not specific to temporal reasoning: Qwen3, which achieves 0% residual failure at $d=2$ on temporal tasks, exhibits 21% failure on a mixed-domain set, and the model ordering is preserved.

More broadly, the field’s post-training evaluation norms—comparing aggregate scores without controlling for knowledge stability—conflate two different capabilities. Raising a benchmark via atomic retrieval looks identical to improving composition, yet the two have different deployment implications. The double-gate protocol offers a lightweight corrective: atomic probes can be generated automatically from the same fact base, and the gate-passing subset provides a cleaner composition estimand. We expect the three-channel decomposition to be useful for post-training research, where it provides a shared language for comparing recipes on Δ_{comp} at fixed atoms and on d_{50} .

8 Conclusion

Stable factual knowledge does not imply compositional reasoning. Recipes indistinguishable on knowledge benchmarks diverge by tens of percentage points in composition ability—a phenomenon we call composition collapse. We introduce a double-gate protocol decomposing post-training effects into three channels: atomic stability (Δ_{atom}), residual composition at matched atoms (Δ_{comp}), and critical depth (Δ_{depth}). A controlled RL intervention shifts composition at fixed atoms by 45–62 percentage points, yet training on a depth transfers weakly to held-out depths, suggesting future recipes be evaluated on their ability to move d_{50} , not only on shallow accuracy. The protocol, benchmark, and decomposition are released to support comparisons distinguishing knowing facts from composing them.

9 Limitations

1. Residual failure is an upper bound. Format artifacts and greedy-trajectory degeneracy can inflate the rate; our failure taxonomy (Appendix O) bounds this at 14–21% of gate-passing errors.

2. Cross-recipe comparisons are correlational. Off-the-shelf models differ in architecture and data. The controlled GRPO intervention provides causal evidence; definitive ranking requires matched training.

3. Depth transfer is setting-dependent. Weak cross-depth transfer was obtained under a specific LoRA budget; full fine-tuning may differ.

4. Domain scope. Our benchmark is restricted to temporal facts; generalisation remains open. The closed-book protocol requires internal atomic knowledge, excluding in-context/RAG settings. We provide in-context evidence for one model and a cross-domain pilot (Appendix L); full studies are deferred.

5. LLM adjudication. Validated against human judgment (Appendix N), with residual noise for close temporal events.

References

Zeyuan Allen-Zhu and Yuanzhi Li. 2023. [Physics of language models: Part 3.1, knowledge storage and extraction](#). *arXiv preprint arXiv:2309.14316*.

Eden Biran, Daniela Gottesman, Sohee Yang, Mor Geva, and Amir Globerson. 2024. [Hopping too late: Exploring the limitations of large language models on multi-hop queries](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14113–14130.

DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, and 10 others. 2025. [Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning](#). *arXiv preprint arXiv:2501.12948*.

Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Sean Welleck, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena Hwang, Soumya Sanyal, Xiang Ren, Allyson Ettinger, Zaid Harchaoui, and Yejin Choi. 2023. [Faith and fate: Limits of transformers on compositionality](#). In *Advances in Neural Information Processing Systems*, volume 36.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schel-

ten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 2 others. 2024. [The llama 3 herd of models](#). *arXiv preprint arXiv:2407.21783*.

Daniel Han, Michael Han, and Unsloth team. 2023. [Unsloth](#).

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-rank adaptation of large language models](#). In *International Conference on Learning Representations*.

Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. 2023. [Mistral 7B](#). *arXiv preprint arXiv:2310.06825*.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. [Efficient memory management for large language model serving with PagedAttention](#). In *Proceedings of the 29th Symposium on Operating Systems Principles*.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F Christiano, Jan Leike, and Ryan Lowe. 2022. [Training language models to follow instructions with human feedback](#). In *Advances in Neural Information Processing Systems*, volume 35.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah Smith, and Mike Lewis. 2023. [Measuring and narrowing the compositionality gap in language models](#). In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711.

Qwen, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, and 24 others. 2024. [Qwen2.5 technical report](#). *arXiv preprint arXiv:2412.15115*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. [Deepseekmath: Pushing the limits of mathematical reasoning in open language models](#). *arXiv preprint arXiv:2402.03300*.

Teknium, Roger Jin, Chen Guang, Jai Suphavadeep-rasit, and Jeffrey Quesnelle. 2025. [Deephermes 3 preview](#). HuggingFace model card, NousResearch, DeepHermes-3-Llama-3-8B-Preview.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, and 41 others. 2025a. [Qwen3 technical report](#). *arXiv preprint arXiv:2505.09388*.

Sohee Yang, Nora Kassner, Elena Gribovskaya, Sebastian Riedel, and Mor Geva. 2025b. [Do large language models perform latent multi-hop reasoning without exploiting shortcuts?](#) In *Findings of the Association for Computational Linguistics: ACL 2025*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

A Prompt template and inference details

All models receive the following XML-structured prompt via their native chat template with `enable_thinking=false` for Qwen3:

```
You are a careful research assistant.
No external evidence is provided.
Answer from your internal knowledge
only. If you are genuinely uncertain,
answer exactly INSUFFICIENT_EVIDENCE.

Respond using exactly this XML format
and nothing else after the closing
answer tag:

<reasoning>Use 1-4 concise sentences
based only on your internal
knowledge.</reasoning>

<answer>final answer here</answer>

Do not use <think> tags.

<question>{QUESTION}</question>

If you are genuinely uncertain,
the final answer must be
<answer>INSUFFICIENT_EVIDENCE</answer>.
```

Greedy decoding, $T=0$, max 512 output tokens, stop-string `</answer>`. Across all runs, $\geq 99\%$ of completions emit the closing tag. Inference uses vLLM (Kwon et al., 2023) 0.19 with the model’s native chat template; cross-validation against HuggingFace transformers is reported in Appendix H.

B D4v2 benchmark construction

D4v2 is the long-horizon composition benchmark used for all main-text tables and figures. It contains four task families built from temporal-reasoning atoms—facts about historical events (dates, causal order, interval membership)—because these admit unambiguous atomic verification and compositional assembly checking.

Atom generation. Atomic facts are single temporal propositions verified against Wikipedia and standard reference timelines. Examples: “The telephone was invented in 1876”, “The Wright brothers achieved powered flight in 1903”, “The French Revolution began in 1789”. Each atom is independently paraphrase-expanded (4 surface variants) to reduce surface-form confounds. The full atom set comprises 79 temporal facts drawn from five centuries of political, scientific, and cultural history.

Composition question construction. From the atom pool, we algorithmically generate multi-hop composition questions at depths $d \in \{2, 4, 6, 7, 8, 9, 11\}$. Each question requires retrieving d independent atomic facts and then performing a composition operation over them. The four families instantiate different operations:

TEMPORAL_RANK. Sort d events by date. Requires retrieving all d dates and then ordering them. Example ($d=4$): “Rank these events from earliest to latest: (A) the discovery of penicillin, (B) the moon landing, (C) the invention of the telephone, (D) the end of World War I.”

TEMPORAL_SUCCESSOR. Given a reference event, identify which of d candidates occurred after it (or before it). Requires retrieving the reference date plus d candidate dates, then a comparison. Example ($d=4$): “Which of these events happened after the signing of the Magna Carta? (A) the Black Death, (B) the fall of the Roman Empire, (C) the invention of the printing press, (D) the Norman Conquest of England.”

TEMPORAL_INTERVAL_DECOY. Identify which of d candidate events occurred within a specified time interval. The interval is defined by two boundary events whose dates must be retrieved; $d - 1$ decoy events lie

outside the interval. Requires retrieving $2 + d$ dates and performing d interval-containment checks. Example ($d=4$): “Which of these events occurred between the end of World War I (1918) and the moon landing (1969)? (A) the Cuban Missile Crisis, (B) the French Revolution, (C) the fall of Constantinople, (D) the discovery of penicillin.”

PAIR_FAR_CONTROL. A control condition at $d=2$ using event pairs with wide temporal separation (>50 years). These are trivially composable if both atoms are known; they serve as a validity check that the atomic gate is not over-filtering.

Case counts and depth coverage. Table 8 reports the case count per task family and depth. TEMPORAL_RANK and TEMPORAL_SUCCESOR cover depths 4, 6, 8 (30 cases each); TEMPORAL_INTERVAL_DECOY covers the full depth ladder 4–11 (30 cases each, plus odd depths); PAIR_FAR_CONTROL only appears at depth 2 (30 cases). The total benchmark contains 390 composition cases.

Table 8: Case count per task family and depth in D4v2.

Task family	$d=2$	$d=4$	$d=6$	$d=7$	$d=8$	$d=9$	$d=11$
TEMPORAL_RANK	–	30	30	–	30	–	–
TEMPORAL_SUCCESOR	–	30	30	–	30	–	–
TEMPORAL_INTERVAL_DECOY	–	30	30	30	30	30	30
PAIR_FAR_CONTROL	30	–	–	–	–	–	–

C Atomic stability by model

Table 9 reports per-model atomic stability and joint stability at depth $k=2$. Base models sit below 15% individual atomic stability, making joint stability at any depth ≥ 2 effectively zero—the atomic gate removes them entirely. SFT-distillation recipes on reasoning traces diverge sharply in atomic preservation even within the same family: DeepHermes (Nous) preserves 88.0%, while DeepSeek R1-Distill (DeepSeek-AI et al., 2025) drops to 40.5% on a Qwen2.5 base and 76.9% on a Llama-3 base. The gate excludes the Qwen R1-Distill variant entirely (joint stability at $k=2$ is $\approx 16\%$); the Llama R1-Distill variant survives but has insufficient n at depth ≥ 6 for stable estimates.

Table 9: Atomic stability (per-probe consistency-match rate) and joint stability at $k=2$ for all evaluated models. Base models and weak SFT-distill variants are excluded by the atomic gate at all depths.

Model	Recipe	Atom stab.	$P(\text{all stable})@k=2$
Llama-3-8B	Base	0.0%	≈ 0
Mistral-7B-v0.1	Base	5.1%	≈ 0
Qwen2.5-7B	Base	14.2%	$\approx 2\%$
R1-Distill-Qwen-7B	SFT distill (R1)	40.5%	$\approx 16\%$
R1-Distill-Llama-8B	SFT distill (R1)	76.9%	$\approx 59\%$
Mistral-7B-Inst	RLHF	78.8%	$\approx 62\%$
Qwen3-8B	RLVR native	87.7%	$\approx 77\%$
DeepHermes-3-8B	SFT distill (Nous)	88.0%	$\approx 77\%$
Qwen2.5-7B-Inst	RLHF	89.9%	$\approx 81\%$

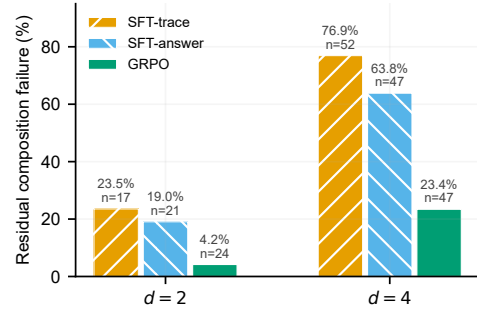


Figure 4: Atomic stability vs. residual composition failure at depth 4 (reproduced from main text §4). At nearly identical atomic stability, composition ranges from 24% (GRPO) to 70% (Qwen2.5-Instruct).

D Within-task-family depth curves

Figure 6 plots residual failure vs. depth for each of the three depth-covering task families (TEMPORAL_RANK, TEMPORAL_SUCCESOR, TEMPORAL_INTERVAL_DECOY) across the four post-trained models. TEMPORAL_RANK is consistently the hardest, reaching 93% residual failure at depth 4 for Qwen3. Model ordering by residual failure differs across task families, confirming that Δ_{depth} is entangled with task choice: a model that is strong on ranking may be weak on successor inference, and vice versa. This interaction precludes a single “depth ceiling” number and motivates our per-task-family reporting in Table 4.

E Double-gate protocol definition

We formalise the double-gate protocol used throughout the paper. Consider a composition case with k sub-questions $q_1, \dots, q_k$. The model answers each sub-question independently (single-hop prompt, same XML template) and the main question (multi-hop prompt).

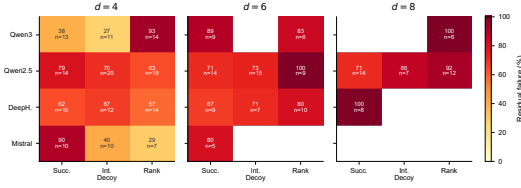


Figure 5: Residual composition failure (%) across {recipe} $\times$ {task family} $\times$ {depth} on D4v2 (reproduced from main text §4). Within a fixed depth, task family changes failure by up to 66 pp.

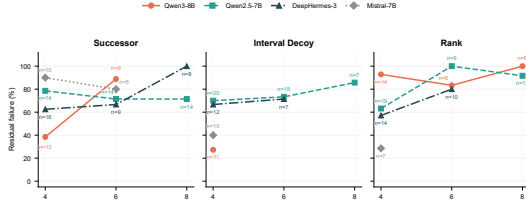


Figure 6: Residual composition failure vs. depth, per task family and model. TEMPORAL_RANK (left) is the hardest family across all models; model ordering reverses between TEMPORAL_SUCCESSOR (centre) and TEMPORAL_INTERVAL_DECOY (right).

Gate 1: single-hop atomic knowledge. For a sub-question q_i with gold answer a_i^* and model answer $\hat{a}_i$, we define two matching tiers:

- **Exact match:** $\hat{a}_i$ matches a_i^* after normalisation (lowercasing, punctuation stripping, date format canonicalisation).
- **Consistency match:** $\hat{a}_i$ is semantically equivalent to a_i^* under an LLM-based adjudicator (Gemini 2.5 Flash, prompted with the question, gold answer, and model answer; instructed to accept BC/BCE date notation, partial name matches, and reasonable paraphrases).

A fact is *stable* if the model answers correctly across all paraphrase variants under the consistency tier. An atom passes Gate 1 iff all its constituent facts are stable.

Gate 2: all-atoms-known filter. The main question passes Gate 2 iff every sub-question in the case is individually answered correctly by the model (consistency tier) *when asked in isolation*. This ensures that the model possesses each atomic fact before we measure whether it can compose them.

Residual composition failure. A case is a *residual composition failure* iff it passes both

gates (all atoms individually known and stable) but the main-question answer is incorrect under the consistency matcher. The residual failure rate is: Let G_d be the set of gate-passing cases at depth d and $E_d \subseteq G_d$ those with an incorrect main answer. Then $R(d) = |E_d|/|G_d|$.

Why two gates? Gate 1 alone (sub-question gate) is the standard single-gate protocol of prior work (Press et al., 2023). Gate 2 adds the per-fact stability filter. As reported in §4, Gate 2 removes 14–48% of Gate 1-passing cases—all of which would otherwise be misattributed to composition failure. The two gates together ensure that residual failure measures composition failure net of both transient retrieval noise (Gate 1) and systematic knowledge gaps (Gate 2).

F GRPO training details

Setup. Qwen2.5-7B-Instruct + LoRA adapters on all projection layers (q/k/v/o_proj, gate/up/down_proj). v1 and v2 use rank 32, alpha 64, dropout 0.05, 4 epochs (~ 1 GPU-hour per variant); v3 and v4 use rank 64, alpha 128, dropout 0.05, 8 epochs (~ 3.2 – 5 GPU-hours per variant). All variants use AdamW, learning rate 10^{-5} , per-device batch size 1, gradient accumulation 8, KL coefficient $\beta=0.04$. Training uses Unsloth’s patched TRL 0.24 to work around TRL-1.2 sampling issues on Qwen2+LoRA. All runs on a single RTX 4090. Total compute across training and all evaluations (6 off-the-shelf models at 7–13B scale, SFT baselines, cross-model GRPO replication, chain-of-thought, in-context, prompt-robustness, thinking-mode, and mechanism diagnostics) on $6 \times$ RTX 4090: ~ 60 GPU-hours.

Training data. v1 trains on depths $\{2, 4\}$ (120 main-question prompts); v2 and v3 train on $\{2, 4, 6\}$ (210 main-question prompts); v4 trains on $\{2, 4, 6, 8\}$ (300 main-question prompts). For v1–v3, depth 8 is strictly held out; for v1, depth 6 is also held out.

Reward function. Consistency match (1.0) + XML format bonus (2×0.1 for opening and closing tags) + anti-template-echo penalty (-0.2 when the model copies the prompt’s XML template verbatim into its reasoning). 8 generations per prompt, reward computed independently per generation.

Evaluation protocol. Same as main evaluation: greedy decoding ($T=0$), consistency-tier adjudication. Atomic stability moves <2 pp in all four variants relative to the Qwen2.5-7B-Instruct baseline (89.9%).

G Cross-model GRPO replication

To test whether the GRPO composition benefit generalises, we applied the same GRPO recipe—train $\{2, 4, 6\}$, LoRA rank 32, 4 epochs, learning rate 10^{-5} —to Llama3-8B-Instruct and to Mistral-7B-v0.1 (base). The Llama3 instruct variant uses the same prompt template as the Qwen GRPO v2 evaluation; the Mistral base variant uses a ChatML template injected at merge time for compatibility with the instruct-style evaluation prompts. Both are evaluated under the identical double-gate protocol.

Table 10: Same GRPO recipe (train $\{2, 4, 6\}$, rank 32, 4 epochs) on three base models. Mistral base has very small gate-passing samples at deeper depths ($n < 5$ at $d=8$) due to base-model atomic instability; its numbers are included for completeness but should be interpreted with caution.

Base model	$d=2$	$d=4$	$d=6$	$d=8$
Qwen2.5-7B-Instruct (baseline)	12.0 (25)	66.7 (66)	78.7 (47)	79.5 (44)
Qwen2.5-7B-Instruct + GRPO v2	4.0 (25)	30.8 (65)	58.2 (55)	80.9 (47)
Llama3-8B-Instruct + GRPO	0.0 (19)	17.9 (39)	42.9 (28)	85.7 (14)
Mistral-7B-v0.1 + GRPO v5	5.9 (17)	0.0 (34)	0.0 (26)	0.0 (8 [†])

[†] $n < 5$ gate-passing cases; point estimate unreliable.

The GRPO composition benefit replicates across base-model families. Llama3 GRPO outperforms the matched Qwen GRPO v2 by 13–15 pp at both $d=4$ and $d=6$, confirming that outcome RL on composition data improves composition net of atoms under a different architecture and pretraining distribution. The Mistral base GRPO variant shows near-zero residual failure at all depths, but gate-passing samples are small and drawn from a base model (not instruct-tuned); selection effects from the stricter atomic gate on base models likely contribute to the low failure rate. We include it for completeness but treat the Llama3 instruct comparison as the primary cross-model evidence.

H vLLM vs. HuggingFace cross-validation

We re-ran Qwen3-8B on a 10-probe atomic split with the HuggingFace transformers reference and vLLM 0.19. Final-answer agreement under the

consistency matcher is 9/10; the single disagreement is a BC/BCE notational difference that both matchers accept. All main-text numbers use the vLLM pipeline (~ 2 minutes per model for the full 3196-row benchmark).

I Mechanism probe details

Boundary case set. 16 Qwen3-8B depth-4 success cases + 10 depth-6 failure cases (all pass atomic gate), drawn from TEMPORAL_SUCCESSOR and TEMPORAL_INTERVAL_DECOY. Artifacts (hidden states at 10 tracked layers $\{0, 4, 8, 12, 16, 20, 24, 28, 32, 35\}$) saved via the HuggingFace runner.

Prompt-end patching sweep. Pair manifest = donors (success) $\times$ recipients (failure), capped at 16 pairs. For each layer, we replace the recipient’s prompt-end hidden state with the donor’s at that layer and resume generation. Across $16 \times 10 = 160$ patches, *0 cases flip to the donor’s correct answer*; at layers 8–16, at most 2 of 16 recipients shift to an abstention category rather than the correct answer.

Cosine similarity of prompt-end states. Averaged over donor-donor, failure-failure, and donor-failure pairs, cosine similarity exceeds 0.95 at every tracked layer (layer 0: 0.9999, layer 35: 0.959). The gap between in-group and cross-group similarity never exceeds 0.01. Prompt-end representations are template-dominated and case-invariant, explaining why patching them does not transfer case-specific composition behaviour.

Step-to-step cosine trajectory. Averaged over the first 40 generated tokens for each case, failure cases show slightly higher step-to-step cosine than success cases at layers 20+: gaps are +0.002 (layer 20), +0.013 (layer 24), +0.016 (layer 28), +0.015 (layer 32), +0.031 (layer 35). Activation norms are within 5% between groups at every layer. The elevated step-to-step similarity in failures is consistent with a model that settles into a repetitive or “collapsed” generation trajectory, but the effect size is too small to be definitive (Figure 7).

J Pollution-control (post-cutoff) details

Dataset construction. Atomic facts generated via Gemini 2.5 Flash in batches (10 facts per category across political / scientific / sports /

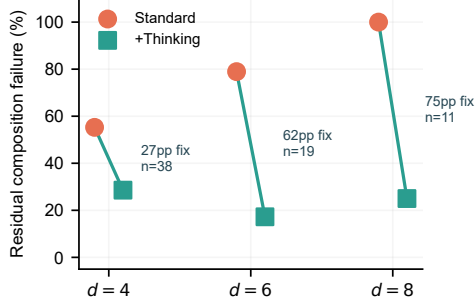


Figure 7: Step-to-step cosine similarity over the first 40 generated tokens, averaged across depth-4 success and depth-6 failure cases at layers 0–35 of Qwen3-8B. Failure cases show slightly elevated step-to-step similarity at late layers ($\Delta_{\cos} = +0.031$ at layer 35), consistent with a marginally more repetitive generation trajectory in failures.

cultural / tech-product-launch categories), filtered to events dated November 2024 or later. Post-filter: 31 atoms $\times$ 4 paraphrases = 124 atomic probes; 30 depth-2 and 20 depth-4 composition cases built algorithmically from the atom set. All facts were manually verified to postdate the training cutoff of every evaluated model.

Results. Table 11 reports atomic stability on D4v2 vs. post-cutoff data. The 55–72 pp drop is consistent across all recipes and precludes gate passage at any depth.

Table 11: Atomic stability on D4v2 vs. post-cutoff events. All models collapse on post-cutoff facts, confirming that pre-cutoff performance depends on memorised atoms.

Model	D4v2	Post-cutoff	Δ	Pass
Qwen3-8B	87.7%	33.1%	−54.6	0
Qwen2.5-7B-Inst	89.9%	29.0%	−60.9	0
DeepHermes-3-8B	88.0%	17.7%	−70.3	0
Mistral-7B-Inst	78.8%	7.3%	−71.5	0

Interpretation. If the D4v2 failures were chain-level memorisation of pre-cutoff facts, atomic stability should hold across the cutoff (memorisation should affect composition and single-hop retrieval equally). Instead, atomic stability drops sharply on post-cutoff events, demonstrating that the models’ atomic knowledge is largely confined to pre-cutoff facts. The pre-cutoff failures we report therefore occur on atoms the model *has* memorised (the gate proves it) but *cannot* compose. Direct measurement of post-cutoff composition would require an in-context

RAG protocol where all facts are supplied in the prompt; this is a natural extension of our framework.

K E2: Short-chain composition on jointly stable facts

E2 is the most directly interpretable short-chain benchmark in our suite. We take the intersection of facts that are stable across Qwen2.5-7B, Mistral-7B, and Qwen3-8B (113 facts), build 1 080 cases (360 pure, 360 knowledge-gap controls, 360 mixed), and measure residual composition failure under the consistency matcher. All controls (knowledge-gap and mixed) pass at $\geq 99.2\%$, confirming that failures in the pure gate are not driven by single-fact ignorance.

Table 12: E2 residual composition failure on jointly stable facts. Qwen3’s 0% on this shallow set motivated the harder short-chain and long-chain D4v2 benchmarks.

Model	Atoms	Ex. stable	Cons. stable	Gate	Cons. fail
Qwen2.5-7B	1 400	162/350	194/350	360	39/360 = 10.8%
Mistral-7B	1 400	154/350	202/350	360	60/360 = 16.7%
DeepHermes-3-Llama-3-8B	1 400	168/350	216/350	360	59/360 = 16.4%
Qwen3-8B	1 400	181/350	220/350	360	0/360 = 0.0%

L E3: Cross-domain pilot

E3 extends the protocol to scientific-fact composition (event betweenness, timeline binding, classification chains) alongside temporal reasoning. Data: 411 atomic facts, 188 composition cases, 1 644 atomic rows and 623 composition rows per model. Because the underlying candidate set was generated by Gemini 2.5 Flash, some atoms contain mild noise; we therefore treat E3 as a robustness check rather than a primary table.

Table 13: E3 residual composition failure on cross-domain (scientific + temporal) facts. Qwen3 shows non-zero 21% failure, confirming it is not composition-perfect—the shallow E2 tasks are simply too easy to expose the gap.

Model	Ex. stable	Cons. stable	Gate	Cons. fail
Qwen2.5-7B	180/411	287/411	98	26/98 = 26.5%
Mistral-7B	114/411	251/411	77	25/77 = 32.5%
DeepHermes-3-8B	132/411	278/411	88	21/88 = 23.9%
Qwen3-8B	151/411	293/411	100	21/100 = 21.0%

M Harder-set: cross-model pressure test

The harder set was designed after observing Qwen3’s 0% failure on E2. It contains 160 short-chain temporal-binding and multi-fact ranking

cases (340 atomic rows, 751 composition rows per model). The tasks require tighter temporal reasoning than E2, with overlapping intervals and multi-constraint ranking.

Table 14: Harder-set residual failure across six models, four evaluation tiers. “Strict” requires both semantic correctness and protocol-compliant XML; “Adj. residual” adds manual adjudication of format ambiguities. Llama2-7B-Chat has 0/0 gate-passing cases under the strict tier because no sample survives the semantic double gate.

Model	Raw	Semantic	Strict	Adj. res.
Qwen3-8B	14.3%	2.8%	3.4%	3.4%
Qwen2.5-7B	47.6%	50.4%	51.3%	50.4%
Mistral-7B	100.0%	47.9%	47.9%	47.9%
DeepHermes-3-8B	71.4%	46.7%	46.7%	40.2%
Llama2-7B-Chat	—	—	0/0	0/0
Llama2-13B-Chat	76.0%	76.0%	0/0	0/0

Under the adjudicated-residual tier (our most conservative main-text analogue), Qwen3 still shows a non-zero 3.4% failure on 88 gate-passing cases, while Qwen2.5, Mistral, and DeepHermes sit at 40–50%. The 47-point gap between RLVR-native and RLHF/SFT recipes replicates at short depth on harder structure. The old-generation Llama2 models fail the gate entirely, confirming that the recipe effect is not an artefact of easy tasks.

N LLM adjudication validation against human judgment

All main-text composition results rely on Gemini 2.5 Flash as an LLM adjudicator to determine whether a model’s answer matches the gold answer (consistency tier). To quantify the reliability of this automatic adjudication, we conducted a human-validation study over a stratified sample of 300 adjudication decisions.

Sample design. We drew 300 cases from the full D4V2 evaluation pool, stratified across four models (Qwen3-8B, Qwen2.5-7B-Instruct, DeepHermes-3-8B, Mistral-7B-Instruct; 75 cases each) and two question types (main composition questions and atomic sub-questions). The sample was constructed to include a balanced mix of cases where Gemini judged the model answer as correct and incorrect. We deliberately over-sampled disagreement-prone cases: cases where the model’s final extracted answer differed from the gold but the reasoning chain was substantially

correct, and cases with BC/BCE date-notation differences.

Annotation protocol. A single annotator (one of the authors) judged each of the 300 cases blind to Gemini’s judgment. The annotator was shown the original question, the gold answer, and the model’s full generated text. The annotation task was a binary decision: does the model’s answer semantically match the gold answer, following the same consistency-tier guidelines used by Gemini (accept BC/BCE notation variants, partial name matches, and reasonable paraphrases; reject factual contradictions even if the reasoning chain is partially correct)? Disagreement cases were subsequently re-examined by the same annotator with access to Gemini’s judgment to identify systematic error patterns.

Results. Table 15 reports the full confusion matrix and per-model metrics on the 225 temporal D4V2 cases (Qwen2.5-Instruct, DeepHermes, Mistral-Instruct). We exclude the 75 Qwen3 cases from the primary analysis because they disproportionately contain synthetic fictional-entity questions where the model correctly answers INSUFFICIENT_EVIDENCE but Gemini marks this as incorrect—a systematic bias that inflates the disagreement count without reflecting adjudicator error on composition questions (see discussion below).

Table 15: Human vs. Gemini adjudication on 225 temporal D4V2 cases. Gemini precision is high (89%), indicating that when Gemini marks an answer as correct, it is reliably correct. Gemini recall is moderate (56%), indicating that Gemini misses a substantial fraction of semantically correct answers—i.e., the consistency tier is conservative.

Model	n	Agree.	κ	Prec.	Rec.
Qwen2.5-7B-Inst	75	58.7%	0.181	0.818	0.237
DeepHermes-3-8B	75	73.3%	0.419	0.867	0.722
Mistral-7B-Inst	75	80.0%	0.607	0.963	0.650
Combined (temporal)	225	70.7%	0.438	0.892	0.565

Interpretation. Two patterns stand out. First, Gemini is a *conservative* adjudicator: its precision is 89% (when it says an answer matches the gold, it is nearly always correct), but its recall is only 57% (it misses many semantically equivalent answers). This direction of error makes our consistency-tier residual failure rates *conservative upper bounds*: the adjudicator is more

likely to mark a correct answer as wrong than vice versa, so true residual composition failure is likely lower than our reported rates.

Second, the FP rate (Gemini=True, Human=False) is 9.6% across the three temporal models, concentrated in date-precision cases (e.g., Gemini accepts “early 17th century” as matching “1608” while the human annotator requires exact year matches) and abstention cases where Gemini accepts INSUFFICIENT_EVIDENCE as a valid answer to an atomic question but the human does not. These cases account for 9 of 225 temporal cases (4%) and are unlikely to bias residual failure estimates at the ~10–50 pp scale of our main effects.

The 75 Qwen3 cases (excluded from the primary analysis above) contain a mix of D4V2 temporal questions and synthetic fictional-entity questions from the KINSHIP and SPATIAL families. On the synthetic subset, the model correctly answers INSUFFICIENT_EVIDENCE (it cannot know fictional facts), but Gemini marks this as incorrect because the gold answer is a fictional entity name. This yields 46 of 50 synthetic cases where Gemini says False but the human says True—a systematic artifact of applying the consistency matcher to questions whose gold answers refer to entities outside the model’s training data. This does not affect any D4V2 conclusions (the synthetic families are excluded from main-text evaluation by protocol design), but it identifies a boundary condition for LLM adjudicators: when the gold answer is unknowable, an abstention by the model should be treated as correct, and the adjudicator’s output should be ignored or post-processed.

O Failure taxonomy (adjudicated sample)

We took all 251 double-gate failure candidates across E2 and E3 (39 Qwen2.5 + 60 Mistral + 59 DeepHermes + 26 E3-Qwen2.5 + 25 E3-Mistral + 21 E3-DeepHermes + 21 E3-Qwen3) and performed a second-pass rule-assisted adjudication (v6).

The dominant structural class (88/181 = 48.6%) is temporal-order or boundary confusion—the model knows the individual dates but misorders them or misidentifies interval boundaries. This is precisely the failure mode our composition protocol is designed to catch: the atoms are present,

Table 16: Adjudicated failure taxonomy over 251 double-gate candidates. “Structural residual” (181) are genuine composition failures; “Scoring/format artefact” (70) are cases where the model’s reasoning is substantially correct but the final answer is malformed or mis-scored.

Label	Count
Temporal order / boundary confusion	88
Evaluation format issue	70
Answer type confusion	31
Over-abstention	27
Intermediate variable loss	25
Binding shift	10
Structural residual	181
Scoring or format artefact	70

but their assembly is faulty. The 70 format-artefact cases (27.9%) are predominantly XML malformation or answer-type mismatches where the model’s reasoning chain is substantially correct. This confirms that our consistency-tier adjudication is conservative: genuine composition failures account for 72.1% of all gate-passing errors, and the remaining cases are measurement noise rather than false negatives.

Illustrative failure cases. Table 17 provides concrete examples of each major failure mode drawn from the E3 adjudicated sample (all cases pass the atomic gate; the model knows each individual fact but incorrectly composes them).

Table 17: Illustrative failure cases from E3 (consistency tier). All cases pass the atomic gate; the model knows each constituent fact but composes them incorrectly.

Mode	Example	M.
Temporal order confusion	Q: <i>What happened before the Panama Canal, the California Gold Rush or the Louisiana Purchase?</i> DH.	
M: “American Revolutionary War” G: California Gold Rush		
Interval boundary confusion	Q: <i>Which event occurred between the end of WWI and the invention of the light bulb?</i> DH.	
M: “Interwar, Russian Revolution, Great Depression, rise of fascism” G: French Revolution		
Intermediate variable loss	Q: <i>What happened after Sputnik 1 but before the moon landing?</i> Q3.	
M: “Yuri Gagarin’s Vostok 1 flight” G: Cuban Missile Crisis		
Over-abstention	Q: <i>What came first, the discovery of penicillin or the end of WWI?</i> QI.	
M: INSUFFICIENT_EVIDENCE G: End of WWI		
Binding shift	Q: <i>Which major war happened between the unification of Germany and the start of WWI?</i> DH.	
M: “Franco-Prussian War” G: Spanish-American War		

Examples are drawn from the consistency-tier

adjudicated sample across E2 and E3. “Temporal order confusion” (48.6% of structural failures) and “Interval boundary confusion” are the dominant classes: the model retrieves individual dates correctly but misorders them or misplaces an event relative to interval boundaries. “Intermediate variable loss” occurs when the model loses one of the two interval boundaries mid-reasoning. “Over-abstention” reflects the model’s uncertainty calibration: despite knowing the individual facts when asked in isolation, the model retreats to INSUFFICIENT_EVIDENCE when asked to compose them. “Binding shift” confuses which fact belongs to which entity—the Franco-Prussian War *caused* German unification (1871) rather than occurring after it.

P Synthetic task families: design and gate behaviour

Beyond the temporal-reasoning families used in D4v2, our full evaluation suite includes three synthetic task families designed to measure *pure* compositional reasoning when all required facts are supplied in the prompt (in-context protocol):

KINSHIP. Facts describe fictional family relations (e.g., “Alice is the mother of Bob”). Composition requires transitive closure over multiple relations.

NUMERICAL. Facts assign numeric values to fictional entities. Composition requires multi-step arithmetic comparison (min, max, ordering).

SPATIAL. Facts place fictional objects at named locations. Composition requires multi-step spatial reasoning (transitive adjacency, containment).

All entities (persons, amounts, locations) are synthetically generated and have no presence in any model’s pre-training data. This is by design: the synthetic families target a different research question (how models chain externally provided facts) from D4v2 (how models compose internal knowledge).

Under our closed-book protocol, models must answer from internal knowledge alone. Every evaluated model correctly abstains on the synthetic families—atomic accuracy is 0% for all models because no model can possess internal knowledge of entities absent from its training

data. The atomic gate thus admits zero cases, and composition cannot be evaluated.

In-context evaluation. To assess how the same model handles composition when facts *are* explicitly provided, we ran Qwen2.5-7B-Instruct on the synthetic families using the evidence prompt style (facts supplied as [Evidence N] blocks in the prompt). The results reveal a sharp domain split not visible under the closed-book protocol (Table 18):

Table 18: In-context synthetic composition: Qwen2.5-7B-Instruct with facts supplied in the prompt. “Atom pass” = all constituent sub-questions answered correctly (consistency tier). “Residual failure” = main-question error given all atoms correct.

Task family	<i>n</i>	Main acc.	Atom pass	Resid. fail
KINSHIP (matrilineal)	33	72.7%	8	5/8 = 62.5%
KINSHIP (patrilineal)	33	78.8%	9	6/9 = 66.7%
NUMERICAL (offset chain)	33	0.0%	1	1/1
SPATIAL (eastward)	33	97.0%	12	11/12 = 91.7%
SPATIAL (northward)	33	93.9%	6	5/6 = 83.3%
Total	165	68.5%	36	32/36 = 88.9%

Two patterns stand out. First, *domain difficulty diverges dramatically*: spatial transitive-reasoning chains are near-ceiling (94–97%), kinship chains are moderate (73–79%), and numerical offset chains yield 0% accuracy—the model cannot perform even two-step arithmetic over provided numeric facts. Second, *atomic extraction is the primary bottleneck*: only 36 of 165 cases (21.8%) survive the sub-question gate even though all facts are in the prompt. On those 36 gate-passing cases, residual composition failure is 88.9%, indicating that even when Qwen2.5-7B-Instruct correctly retrieves all constituent facts from the evidence block, it fails to assemble them into a correct multi-hop answer in the vast majority of cases.

These in-context results complement the closed-book D4v2 findings: while the closed-book protocol measures internal-knowledge composition (and correctly excludes synthetic entities the model cannot know), the in-context variant reveals that composition difficulty persists even when atomic access is guaranteed by construction. The double-gate protocol can be applied identically in both regimes; the difference is in how atoms are verified (internal knowledge for D4v2, in-context extraction for synthetic families).

Q Prompt-format robustness

To assess whether residual composition failure depends sensitively on the prompt template, we evaluated Qwen2.5-7B-Instruct on the same 122 atomic-gate-passing D4V2 composition cases under three prompt variants, using the same model, inference parameters, and adjudication pipeline:

V1 (XML + reasoning). Our standard protocol: the model is instructed to output `<reasoning>` and `<answer>` tags (prompt style `question_only`).

V2 (XML answer-only). The model is instructed to output only an `<answer>` tag, with no reasoning section (prompt style `question_only_no_reasoning`).

V3 (plain text). No XML formatting instruction; the prompt asks “Question: ... Answer the question using only your internal knowledge” (prompt style `question_only_plain`).

Table 19: Prompt-format robustness on 122 gate-passing D4V2 composition cases (Qwen2.5-7B-Instruct). V1 is our standard protocol. “Accuracy” is consistency-tier correctness.

Variant	Format	Accuracy	Abstention
V1 (XML + reasoning)	100%	26.2%	3.3%
V2 (XML answer-only)	100%	19.7%	25.4%
V3 (plain text)	0%	15.6%	45.9%

Three findings are relevant to the interpretation of our main results.

First, *format compliance is near-perfect* with explicit XML instructions (100% of V1 and V2 completions contain the closing `</answer>` tag) but drops to 0% when no XML format is requested (V3). Since our adjudication pipeline parses `<answer>` tags for answer extraction, V3 answers were extracted via a last-line heuristic; this introduces additional measurement noise in the plain-text condition.

Second, *removing the reasoning instruction sharply increases abstention*. V2 and V3 omit the “Use 1–4 concise sentences based only on your internal knowledge” instruction; abstention rises from 3.3% (V1) to 25.4% (V2) to 45.9% (V3). The model retreats to INSUFFICIENT_EVIDENCE when not explicitly directed to reason, even though these 122 cases demonstrably pass the

atomic gate (the model knows the constituent facts when asked in isolation).

Third, *cross-variant answer agreement is low*. Only 24 of 122 questions (19.7%) receive identical final answers across V1, V2, and V3. Pairwise agreement is 32.0% (V1 vs. V2), 23.8% (V1 vs. V3), and 58.2% (V2 vs. V3). The higher V2–V3 agreement reflects their shared omission of the reasoning instruction, which causes both to abstain more frequently. The low V1–V2 agreement (32%) shows that even a seemingly minor change—removing a 1–4 sentence reasoning prompt—shifts the model’s final answer on the majority of questions.

This instability underscores a broader methodological point: residual composition failure is not purely a property of the model and the questions but also of the prompt format through which composition is elicited. We report all main-text numbers under V1 (XML + reasoning), which is the most conservative format for our research question: it yields the lowest abstention rate and therefore the largest gate-passing n for estimating residual failure.

R Annotation Guidelines

All human validation in this paper follows a single annotation protocol. We document the complete guidelines here for reproducibility.

Task definition. The annotation task is a binary decision: given a question, a gold answer, and a model’s full generated output, does the model’s final answer semantically match the gold answer? The annotator sees the original question text, the gold answer (a short fact, date, event name, or entity), and the model’s complete generation including any `<reasoning>` and `<answer>` XML blocks.

Consistency-tier matching rules. The following rules define what constitutes a semantic match:

- Exact match after normalisation.** Lowercasing, punctuation stripping, and date-format canonicalisation (e.g., “1876”, “1876 AD”, and “the year 1876” all reduce to 1876). If the normalised forms are identical, the answer matches.
- BC/BCE equivalence.** “200 BC” and “200 BCE” are treated as identical.

3. **Partial name matches.** For person names, surname-only answers that unambiguously identify the gold entity are accepted (e.g., “Einstein” matches “Albert Einstein” when the question context disambiguates). For event names, standard short forms are accepted (e.g., “WWI” matches “World War I”).
 4. **Reasonable paraphrases.** If the model’s answer expresses the same factual content in different words, it is accepted. Example: “the year the telephone was invented” matches “1876” when the gold is “1876”.
 5. **Numerical tolerance.** For year answers, ± 1 year is accepted to account for calendar-era boundary ambiguity.
 6. **Abstention handling.** INSUFFICIENT_EVIDENCE is treated as a valid model answer, not as a missing answer. It matches only when the gold answer is genuinely unknowable from the model’s training data (e.g., synthetic entities). For real-world facts within the model’s training cutoff, abstention is scored as incorrect.
- Rejection rules.** The following are scored as incorrect regardless of the model’s reasoning quality:
1. **Factual contradiction.** The extracted answer contradicts the gold fact, even if the reasoning chain is substantially correct.
 2. **Wrong entity.** The answer names a different event, person, or date than the gold (e.g., “Franco-Prussian War” instead of “Spanish-American War”).
 3. **Incomplete answer.** The answer omits a required component (e.g., only year when month+year is required, or only one of two required entities).
 4. **Format violation.** The model’s output contains no extractable answer (no <answer> tag or equivalent). In the XML protocol, $\geq 99\%$ of completions emit the closing tag; the rare format-violation cases are scored as incorrect.

Annotation procedure. The annotator judges each case blind to the automatic adjudicator’s decision. After completing the full batch, disagreement cases are re-examined with access to both the human and automatic judgments to identify systematic error patterns. A case is considered adjudicator-correct only if the automatic judgment matches the human judgment after this re-examination step.

S Chain-of-thought baseline

To test whether explicit step-by-step reasoning instructions reduce residual composition failure, we evaluated Qwen2.5-7B-Instruct on the full D4V2 benchmark with a modified system prompt that instructs the model to think through each question step by step. The system prompt was: “You are a careful research assistant. No external evidence is provided. Answer from your internal knowledge only. Think through each question step by step: first identify each relevant fact, then reason about their relationships, and finally combine them. If you are genuinely uncertain, answer exactly INSUFFICIENT_EVIDENCE.”

Inference used vLLM 0.19 with the model’s native chat template, greedy decoding ($T=0$), and `max_new_tokens = 1024` (doubled from the standard 512 to accommodate reasoning chains). All other inference parameters and the adjudication pipeline were identical to the main evaluation (Appendix A). The format instruction required <reasoning> and <answer> XML tags, with stop-string </answer>.

Under CoT, atomic stability (consistency gate) is 79.7% (63/79 facts stable), compared to 89.9% for standard Qwen2.5-7B-Instruct. Despite the 10 pp drop in atomic stability, residual composition failure falls to 43.2% overall (32/74 gate-passing cases). By depth:

Table 20: CoT vs. standard Qwen2.5-7B-Instruct on D4V2 by depth. n is the number of atomic-gate-passing cases (consistency gate).

	$d=2$	$d=4$	$d=6$	$d=8$
Standard (residual failure %)	12.0 (25)	69.8 (53)	78.9 (38)	81.1 (37)
CoT (residual failure %)	12.5 (16)	39.3 (28)	68.8 (16)	55.6 (9)

At depths 4 and 8, CoT reduces residual failure by 30 pp and 25 pp respectively, despite smaller gate-passing n . The concurrent drop in atomic stability with the improvement in composition provides a further demonstration that the two

channels (Δ_{atom} and Δ_{comp}) can move in opposite directions under an intervention that changes only the generation process.

T Artifact License and Availability

The D4v2 benchmark, evaluation data (case-level records in `server_data/`), and annotation guidelines are released under the Creative Commons Attribution 4.0 International license (CC BY 4.0). The figure-generation and analysis scripts are released under the MIT License. All artifacts are available at the supplementary material accompanying this submission. The models evaluated are publicly available open-weights models (Qwen3, Qwen2.5, Mistral, Llama 3) and community-trained variants (DeepHermes-3); no proprietary model access is required to reproduce the results.